

In-course Assessment Examination – 02
CSC1213(P) – Object Oriented Programming
Department of Physical Science
Answer all Questions

Time allowed: 75 Minutes

June 2023

Consider a scenario where you are building a software application to manage a library. The application needs to handle books and their associated information, such as title, author, and availability. Implement the following classes to fulfill the requirements.

Q1.

Create a class called **Book** with the following attributes:

- **title** (String): to store the title of the book.
- **author** (String): to store the name of the book's author.
- **availability** (boolean): to represent whether the book is available for borrowing (true) or not (false).

Include the following methods in the **Book** class

- Constructor to initialize **title, author, availability**
- Provide appropriate getter and setter methods to access and modify the attributes.

Q2.

Create a subclass called **FictionBook** that inherits from the **Book** class.

The **FictionBook** class should have an additional attribute:

- **genre** (String): to store the genre of the fiction book.

Include the following methods in the **FictionBook** class

- Constructor to initialize **title, author, availability** and **genre**
- Provide appropriate getter and setter methods to access and modify the attribute.

Q3.

Create a class called **Library** that represents a collection of books.

The **Library** class should have the following attributes:

- **books** (1D array of **Book** objects): to store the books available in the library.

Include the following methods in the **Library** class

- Constructor **Library(int capacity)**: to initialize **books** with its capacity
- Provide appropriate getter and setter methods to access and modify the attribute.
- **addBook(Book book)**: to add a book to the library
- **removeBook(int index)**: to remove a book from the library of its index value.
- **displayBooks()**: to display the details of all the books in the library.

Q4. Implement the above classes in the Main class and perform the following actions:

1. Create the instances of **FictionBook** with its attributes using the details given below.
fictionBook1 = title: "The Great Gatsby", author: "F. Scott Fitzgerald", availability: true, genre: "Classic"
fictionBook2 = title: "To Kill a Mockingbird", author: "Harper Lee", availability: true, genre: "Drama"
fictionBook3 = title: "1984", author: "George Orwell", availability: true, genre: "Dystopian"
fictionBook4 = title: "Pride and Prejudice", author: "Jane Austen", true, genre: "Romance"
fictionBook5 = title: "The Catcher in the Rye", author: "J.D. Salinger", availability: true, genre: "Coming-of-age"
fictionBook6 = title: "The Hobbit", author: "J.R.R. Tolkien", availability: true, genre: "Fantasy"
fictionBook7 = title: "The Chronicles of Narnia", author: "C.S. Lewis", true, genre: "Fantasy"
fictionBook8 = title: "Harry Potter and the Philosopher's Stone", author: "J.K. Rowling", availability: true, genre: "Fantasy"
2. Create an instance of **Library** with a capacity of 8.
3. Add the created instances of **FictionBook** to the Library using the **addBook** method.
4. Display the details of all the books in the library using the **displayBooks** method.
5. Remove the book **"Rich Dad Poor Dad"** from the library using the **removeBook** method.
6. Remove the book **"1984"** from the library using the **removeBook** method.
7. Display the details of all the remaining books in the library using the **displayBooks** method.
8. Display the details of the all books which comes under **Fantasy** genre.

The output of the program looks like this:

-----All books in the library-----

Title: The Great Gatsby
Author: F. Scott Fitzgerald
Availability: Available

Title: To Kill a Mockingbird
Author: Harper Lee
Availability: Available

Title: 1984
Author: George Orwell
Availability: Available

Title: Pride and Prejudice
Author: Jane Austen
Availability: Available

Title: The Catcher in the Rye
Author: J.D. Salinger
Availability: Available

Title: The Hobbit

Author: J.R.R. Tolkien
Availability: Available

Title: The Chronicles of Narnia
Author: C.S. Lewis
Availability: Available

Title: Harry Potter and the Philosopher's Stone
Author: J.K. Rowling
Availability: Available

-----Fantasy books in the library-----

Title: The Hobbit
Author: J.R.R. Tolkien
Availability: Available

Title: The Chronicles of Narnia
Author: C.S. Lewis
Availability: Available

Title: Harry Potter and the Philosopher's Stone
Author: J.K. Rowling
Availability: Available

-----Removing book Rich Dad Poor Dad from library-----

No such book in library!

-----Removing book 1984 from library-----

Title: 1984
Author: George Orwell
Book removed from library!

-----Remaining books in the library-----

Title: The Great Gatsby
Author: F. Scott Fitzgerald
Availability: Available

Title: To Kill a Mockingbird
Author: Harper Lee
Availability: Available

Title: Pride and Prejudice

Author: Jane Austen
Availability: Available

Title: The Catcher in the Rye
Author: J.D. Salinger
Availability: Available

Title: The Hobbit
Author: J.R.R. Tolkien
Availability: Available

Title: The Chronicles of Narnia
Author: C.S. Lewis
Availability: Available

Title: Harry Potter and the Philosopher's Stone
Author: J.K. Rowling
Availability: Available